UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

a) The Von Neumann Architecture was first introduced inwhich computer generation?

b) Draw the Von Neumann Architecture.

c) Central Processing Unit (CPU) is one of the components of a Von Neumann Architecture whereas Control Unit and Arithmetic Logic Unit are the main components of a CPU. What is the main task of a CU?

QUESTION 2 (5 marks)

List the name of the inventor of the following devices:

a) Mechanical Calculating Machine

b) Difference Machine

c) Analytical Machine

d) IAS Machine

e) Integrated circuit

QUESTION 3 (5 marks)

Differentiate between DRAM and SRAM in term of the following characteristics:

a) Speed.

b) Size per bit.

c) Cost.

d) Type of bits stored.

e) Refreshing need when powered.

QUESTION 4 (5 marks)

a) What is a cache?

b) Describe TWO (2) ways to store data in cache.

QUESTION 5 (10 marks)

a) Describe THREE (3) types of interfaces to external devices.

b) Explain TWO (2) types of interrupt.

QUESTION 6 (10 marks)

Describe FOUR (4) classifications of multiple processor architecture with an example each.

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QUESTION 7 (10 marks)

a) Differentiate between organization and architecture.

b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

QUESTION 8 (10 marks)

a) convert 46.0703'125n to a floating-point number in IEEE single precision format in hexadecimar varue form. show your comprete workflow'

b) Convert 8F080000rr in IEEE single precision floating point number to decimal value form. Show your workflow.

QUESTION 9 (10 marks)

Question 9 is based on Figure 1. Figure 1 illustrates the multiple interrupts that occur in a computer system.

a) Define interrupt.

b) Rank the interrupts based on their priority from the most important to the least important.

c) Rank the interrupts from the most important to the least important if disable interrupt is applied.

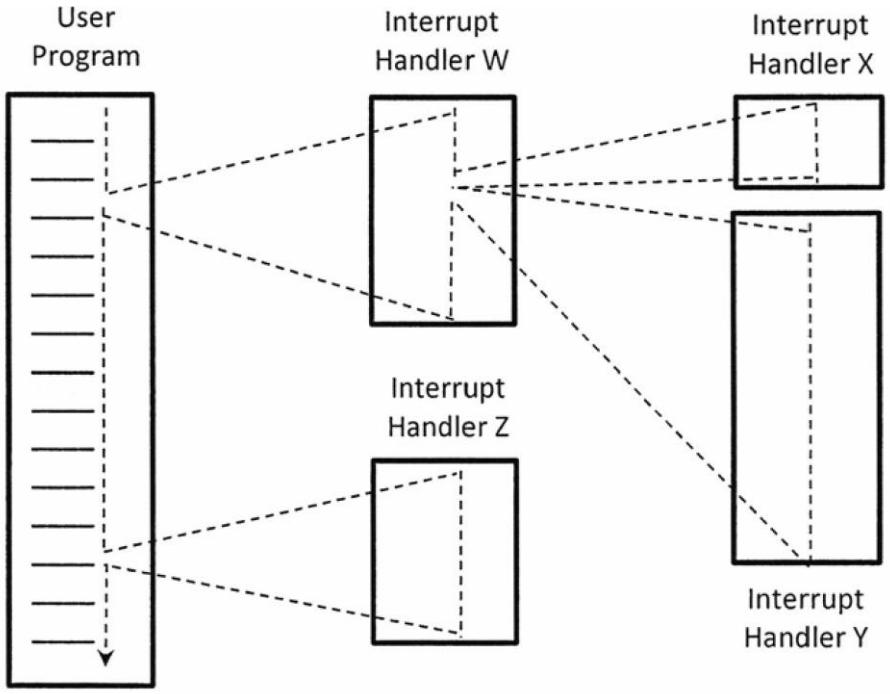


Figure 1: Multiple interrupts

QUESTION 10 (10 marks)

Describe the function of each register in the computer system.

PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 2 and Table 1.
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 2 using one-, two- and three-address instructions. The instructions available for use are provided in Table 1. Note that Register A, B, C, D, E and F cannot be modified and Register X, Y and Z can be used as temporary register. Final answer must be stored in Register X.

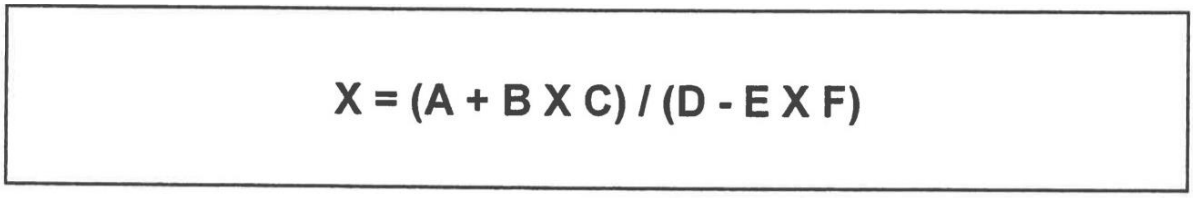


Figure 2

Table 1 : Machine Instruction

i Nia dress		**address**
LOAD M	MOVE (X e Y)	MOVE (X e Y)
STORE M	ADD (X k.. X + Y)	ADD (X e Y + Z)
ADD M	SUB (X e X - Y)	SUB (X e Y-Z)
SUB M	MUL (X e X x Y)	MUL (X e Y x Z)
MUL M	DIV (X e x / Y)	DIV (X e Y / Z)
DIV M		

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